

MIT App Inventor: Beginner's Manual

1. Home Page & Project Dashboard

When you first log in at MIT App Inventor (<https://appinventor.mit.edu>), you land on the Projects Dashboard, where you can create a new app or open an existing one.

The image shows two screenshots of the MIT App Inventor interface. The top screenshot is the homepage, featuring a banner for the 'GLOBAL AI HACKATHON' with a 'Meet the Winners' button. Below the banner are statistics: 'USERS TODAY: 46.69 K', 'USERS THIS MONTH: 2.35 M', 'ALL-TIME USERS: 25.1 M', and 'APPS BUILT: 121.2 M'. The bottom screenshot is the 'Projects Dashboard' with a '+ New project' button highlighted by a red speech bubble that says 'Create a new project'. A 'Welcome to App Inventor!' dialog box is also visible, providing information about updates and device connection.

MIT APP INVENTOR Create Apps!

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GLOBAL AI HACKATHON

Meet the Winners

RAISE Initiative
Responsible AI for Social
Empowerment and Education

App Inventor
Foundation

USERS TODAY: 46.69 K

USERS THIS MONTH: 2.35 M

ALL-TIME USERS: 25.1 M

APPS BUILT: 121.2 M

MIT APP INVENTOR

Connect Build Settings Help English

+ New project Move... Export Gallery Publish

Name Date Created

Create a new project

Welcome to App Inventor!

Welcome to MIT App Inventor

We've Updated the default model used by the Chatbot Component.

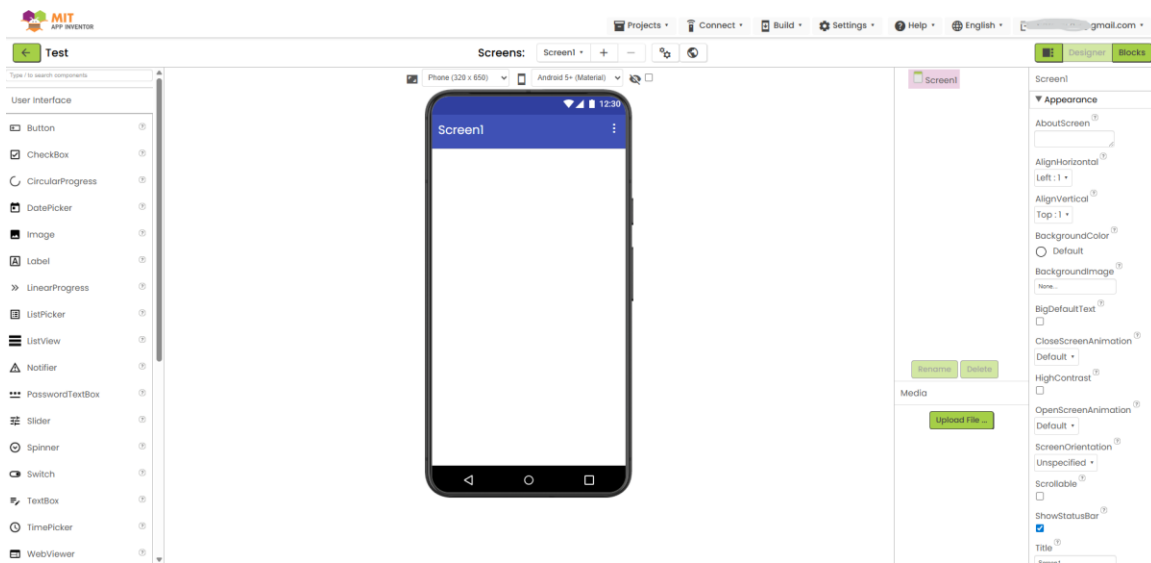
[Read this blog post for more information.](#)

Got an iPhone or Android phone? Find out how to [set up and connect an iOS or Android device.](#)

Continue

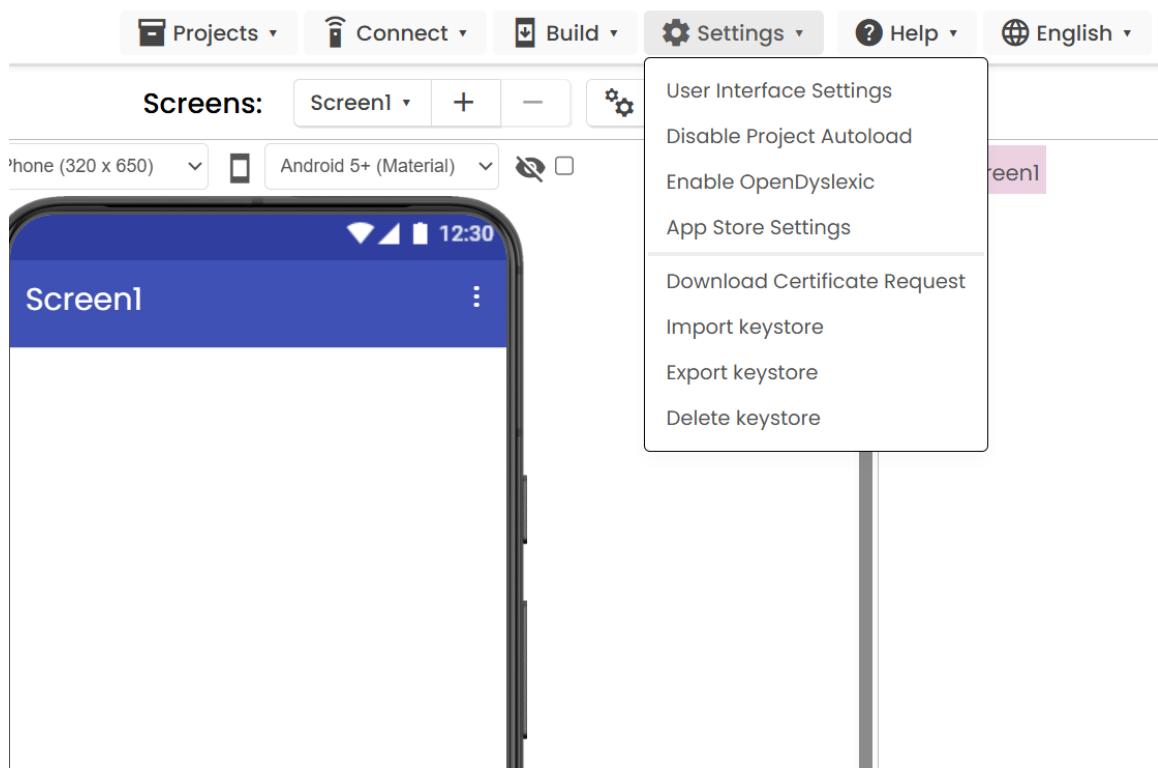
☐ Do Not Show Again

Neo interface

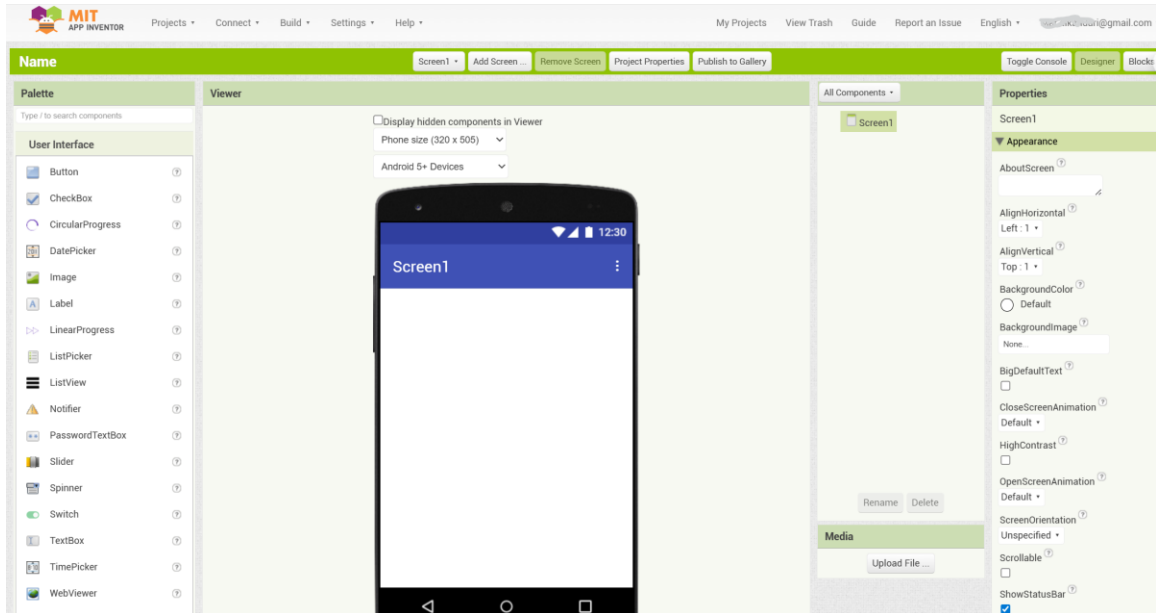
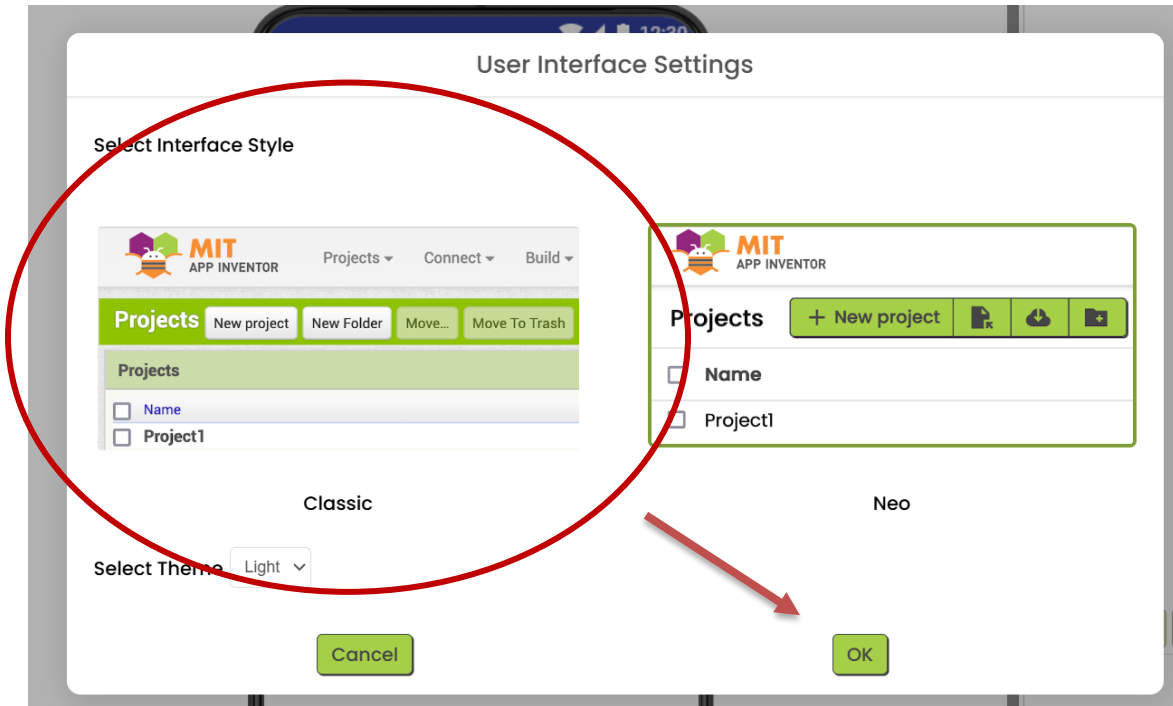


Classical interface

- Go to Settings → select User Interface Settings

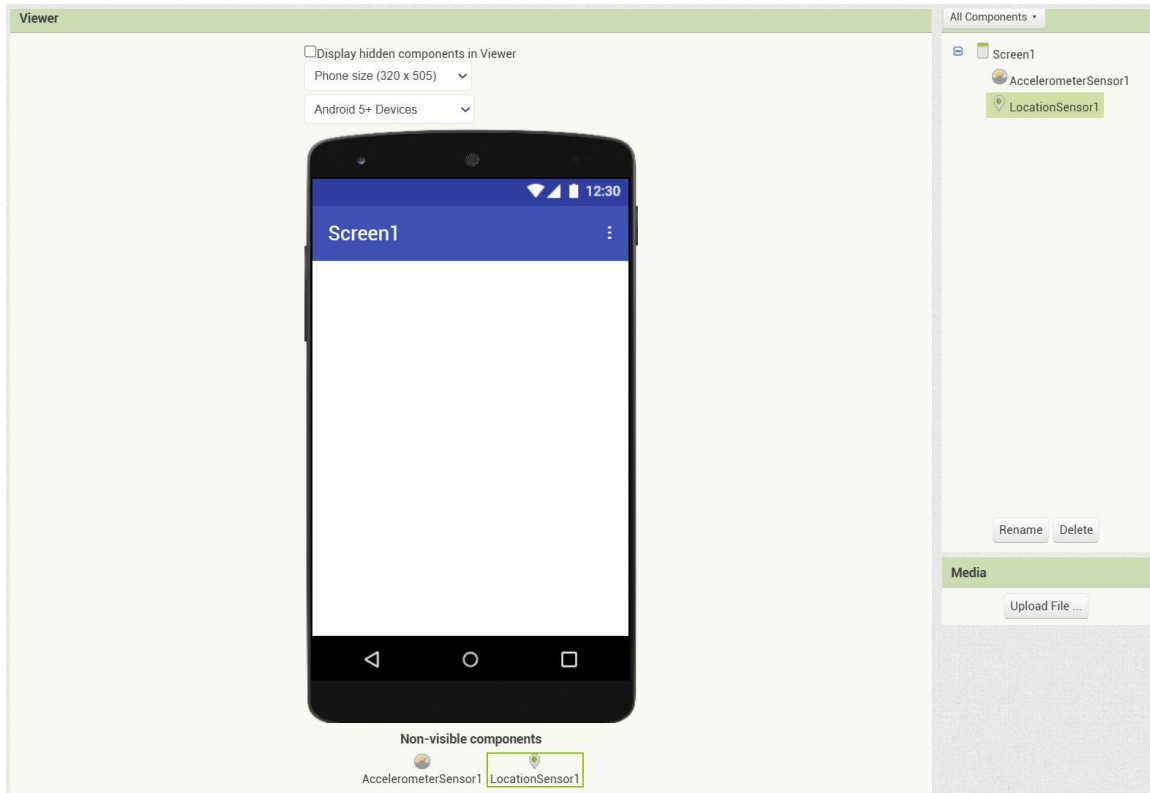


Choose Classical interface style → click OK



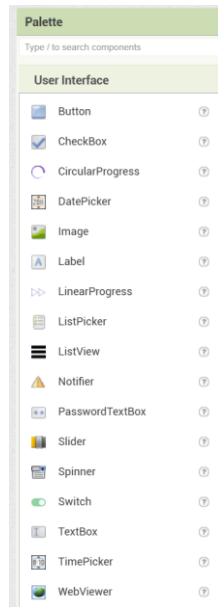
2. Designer View (User Interface Setup)

The Designer is where you create the app's visual interface. You'll see a mock phone screen in the center, a Palette of components on the left, and Properties on the right.



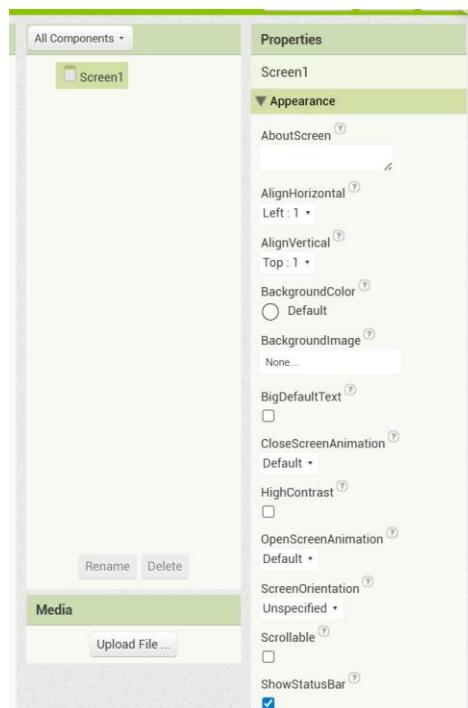
3. Component Palette

The Palette (left side of Designer) contains Visible components (Button, Label, Image, TextBox) and Non-visible components (Sensors, Media, Storage, Connectivity).



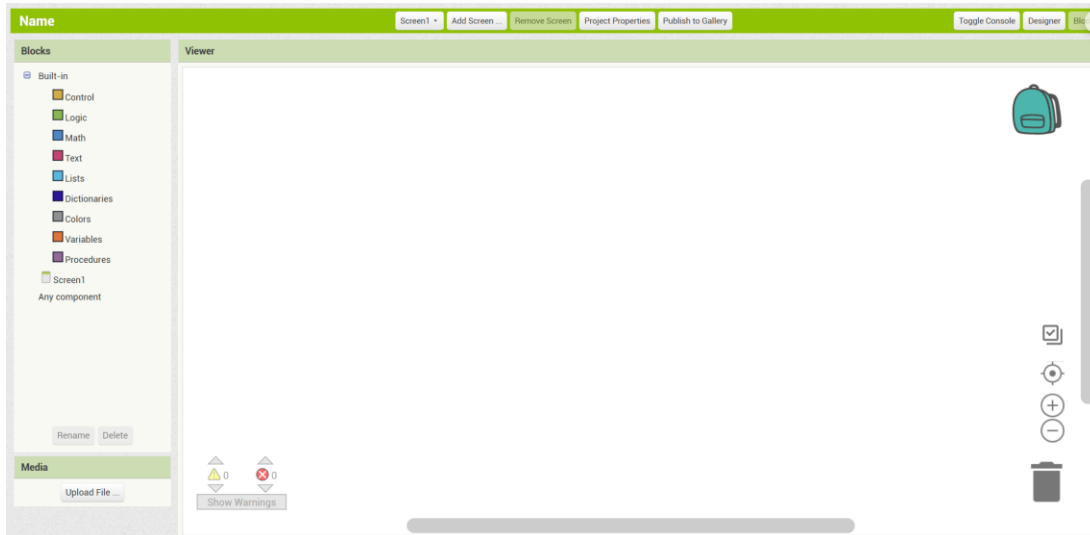
4. Properties Panel

On the right side of the Designer, you'll also find the Properties Panel, where you can customize each component (size, color, font, alignment, etc.).



5. Blocks Editor (Programming the App)

Switching to the Blocks tab lets you add logic using colored code blocks. Blocks snap together like puzzle pieces, so you don't worry about syntax errors.



5.1 Built-in Blocks (Programming Essentials)

In the Blocks Editor, MIT App Inventor provides a set of built-in blocks. These are the foundations of your app's logic. Each category is color-coded to make programming easier. Sample Commands are presented as the following. You can explore full list of blocks via MIT APP Inventor:

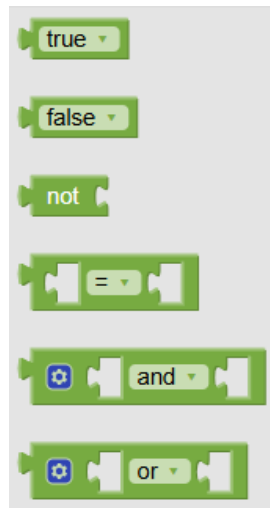
Control

Blocks for decision-making and flow. Includes conditionals (if/then), loops (for each, while), and event controls.



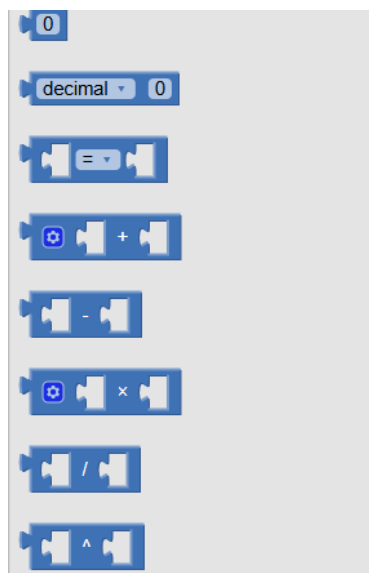
Logic

Handles true/false values, comparisons (e.g., =, <, >), and Boolean operators (and, or, not).



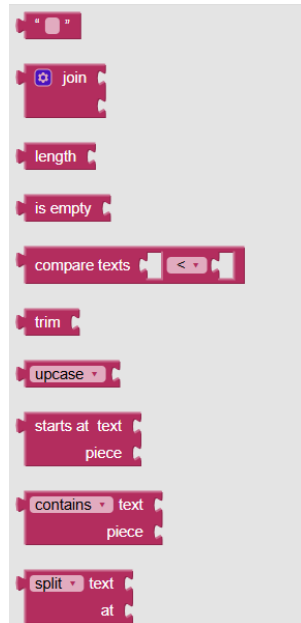
Math

Lets you perform arithmetic operations, random number generation, rounding, and advanced math functions.



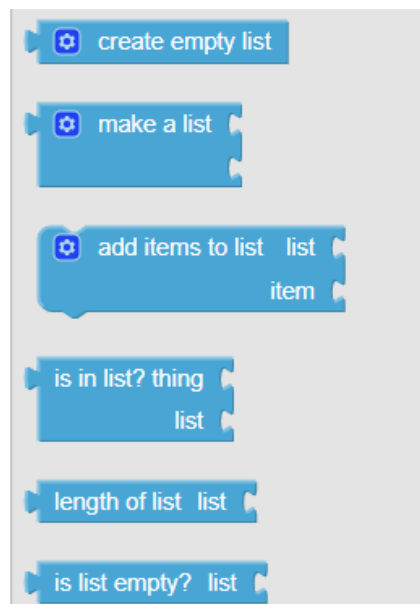
Text

For creating, combining, and manipulating text (strings). Example: joining words, changing case, or extracting parts of a string.



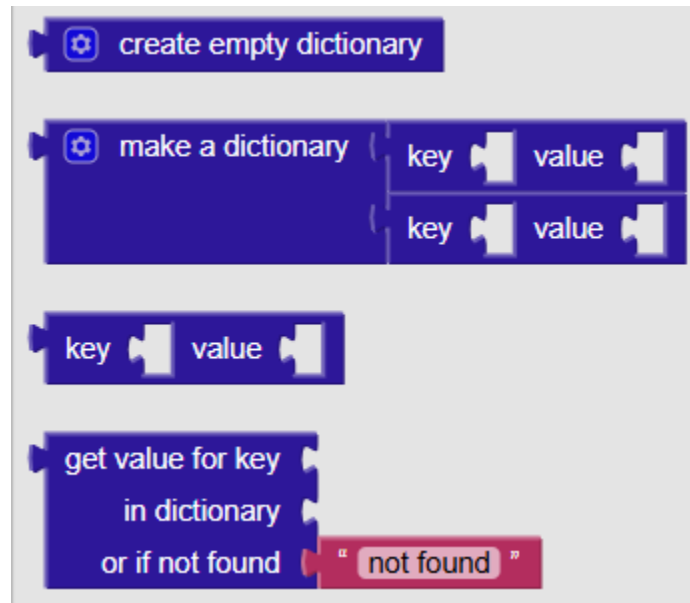
Lists

Allows storing and organizing multiple items in a list. You can add, remove, or search for items.



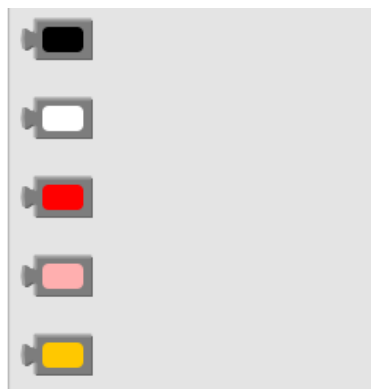
Dictionaries

Works like a map with key-value pairs. Example: storing student: grade pairs.



Colors

Blocks for defining and using colors in your app (RGB values or predefined colors).



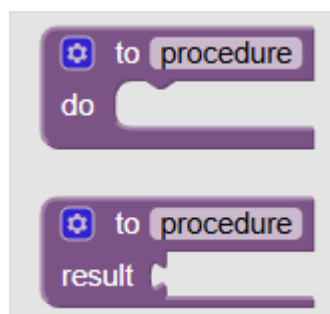
Variables

Used to store information (numbers, text, lists) that can change while the app runs.



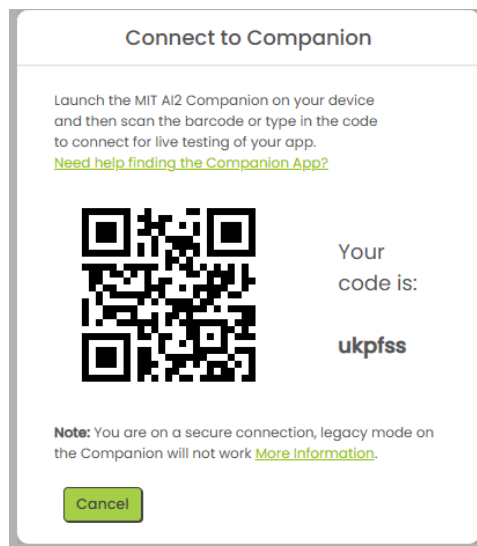
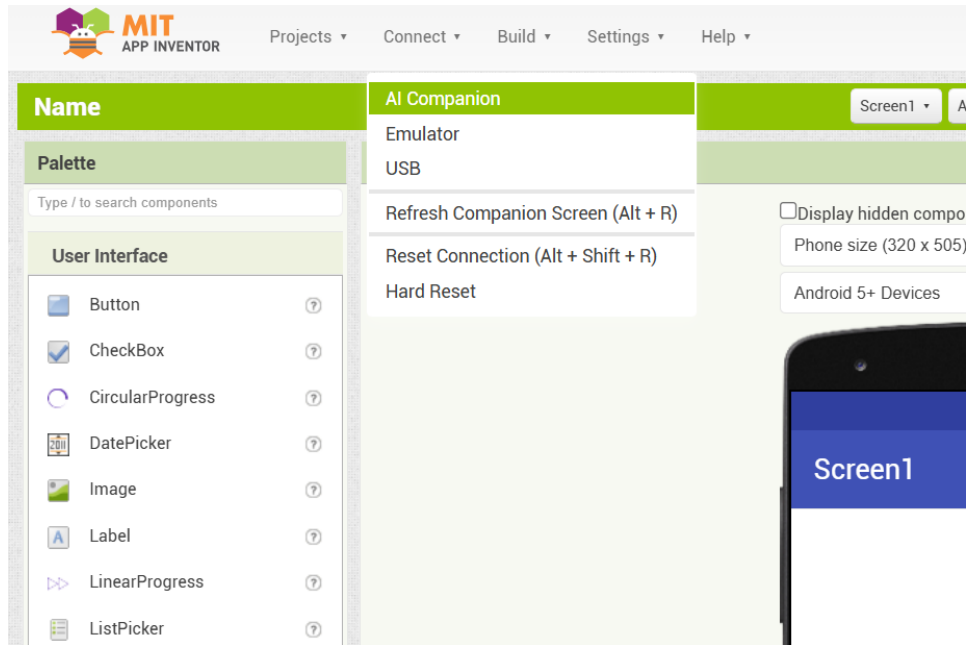
Procedures

Allow you to group blocks into reusable functions, making your app more organized and efficient.



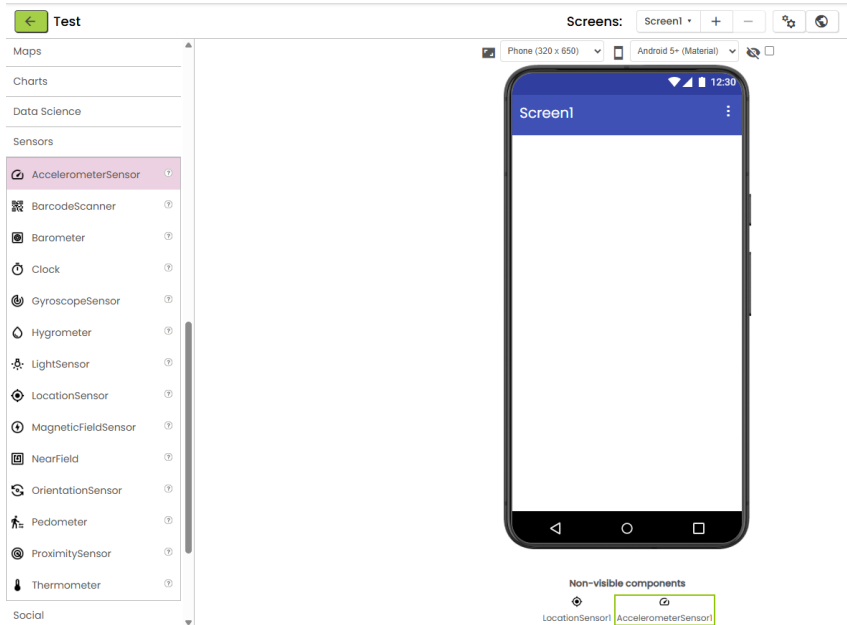
6. AI Companion (Real-Time Testing)

Using the MIT AI2 Companion app, you can scan a QR code to run your app instantly on a phone or emulator.



7. Sensors & Advanced Features

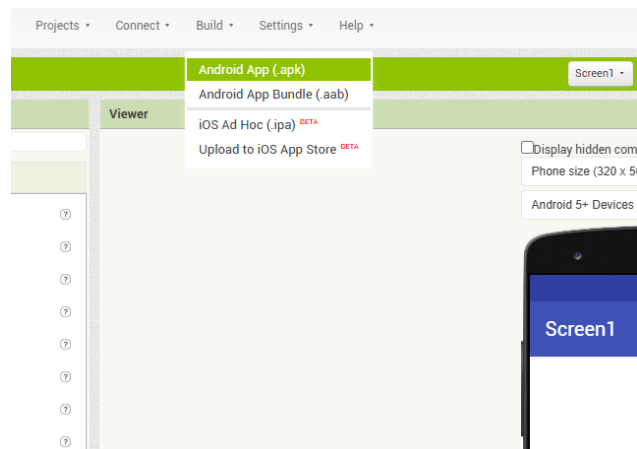
MIT App Inventor includes device features: GPS, accelerometer, camera, Bluetooth, and speech recognition. These allow you to make interactive apps beyond simple buttons.



8. Packaging & Exporting

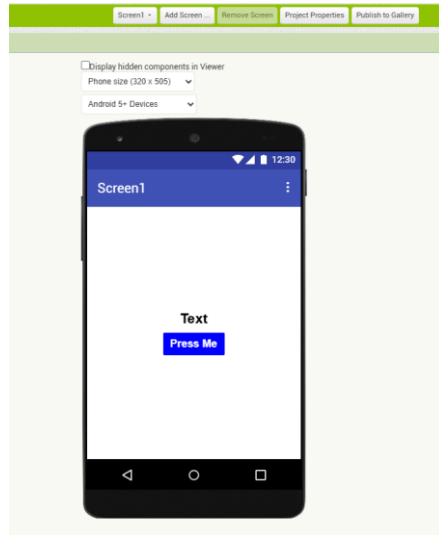
When your app is ready, you can:

- From **Build**: Package as an APK for Android.
- Share via QR code.
- Publish to Google Play.



9. Example: Hello World App

1. In Designer: Drag a Button and a Label.
2. In Blocks: Add logic → When Button.Click → set Label.Text to 'Hello World'.
3. Test with AI Companion.



10. Resources

- MIT App Inventor Official Site: <https://appinventor.mit.edu>
- Beginner Tutorials: <https://appinventor.mit.edu/explore/ai2/tutorials>
- Community Forum: <https://community.appinventor.mit.edu>